Failure-Trace-Guided Gold-Free Policy Gating for Budgeted LLM Reasoning: Cohere Matched-Budget Evidence on Disjoint 720-Example Evaluation

Soroush Vahidi[1*]

[1*]Department of Computer Science, New Jersey Institute of Technology, Newark, NJ, USA.

Corresponding author(s). E-mail(s): sv96@njit.edu;

Abstract

We study *matched-budget answer selection*: given candidate answers already generated by large-language-model methods under the same per-method inference budget, the task is to surface the best answer without additional model calls and without access to ground-truth labels at runtime. We present **FIX-2+FIX-4** (Combined Failure-Trace-Guided Allocator), a deterministic two-gate policy that uses failure-trace metadata logged as a side-effect of the frontier generation run. The first gate, the Low-Depth Guard, fires when the frontier search terminates on a structurally weak single branch—signalled by a logged override-reason metadata field—and substitutes an external majority-vote answer. The second gate, the External-Consensus Fallback, fires when all three independent external baselines unanimously disagree with the frontier's selected answer. Both gates are strictly gold-free: no correctness labels, ground-truth answers, or example identifiers are accessed at decision time.

We evaluate on the GSM8K benchmark [1] using Cohere command-r-plus-08-2024 under a matched per-method budget cap of six logical inference calls per example. On a disjoint 720-example aggregate spanning three independent evaluation seeds, FIX-2+FIX-4 achieves 80.69% accuracy versus 77.64%, 77.08%, and 75.14% for three independently sourced external baselines (L1, S1, and TALE). Source-stratified 95% bootstrap confidence intervals are strictly positive for all four pairwise comparisons, with the lower bound versus the best external at **+1.11** percentage points. All findings are scoped to the tested provider and benchmark; generalization to other settings requires independent replication.

Keywords: Large language models, budgeted inference, answer selection, policy gating, test-time compute

1 Introduction

Scaling inference-time computation offers a practical path to improved accuracy without additional model training [2]. By generating multiple candidate answers under a fixed budget and aggregating or selecting among them, a system can achieve accuracy gains comparable to those from model scaling, at inference time [3]. In settings where the inference budget is strictly matched—every compared method receives the same number of model calls per example—the quality of the final answer depends entirely on the selection rule applied to the already-generated pool. Majority vote treats all candidates equally; but generation runs produce metadata that is richer than raw candidate answers, and this metadata may carry signal about which candidates are more reliable.

We study this question in a concrete, reproducible setting. The task is grade-school arithmetic reasoning on the GSM8K benchmark [1], the provider is Cohere command-r-plus-08-2024, and the budget is $B = 6$ inference calls per example. A frontier generator produces candidate answers via program-aided execution [4]; three independent external methods (L1, S1, and TALE) provide reference answers under the same budget. Given these already-generated outputs, the goal is to select a final answer without additional inference calls and without using gold labels.

The main contribution of this paper is **FIX-2+FIX-4** (Combined Failure-Trace-Guided Allocator), a deterministic two-gate policy that relies on failure-trace metadata logged during normal frontier execution. The *Low-Depth Guard* (FIX-2) fires when the frontier's `override_reason` field signals a structurally weak single-branch output, substituting an external majority vote. The *External-Consensus Fallback* (FIX-4) fires when all three external baselines unanimously prefer a different answer than the frontier's selection. Both gates are strictly gold-free: no correctness labels, ground-truth answers, or example identifiers are accessed at decision time. The two gates operate in strict priority order; at most one fires per example.

On a disjoint Aggregate-720 combining three independent evaluation seeds (seeds 41, 61, 71; $300 + 120 + 300$ examples with zero example-id or question-hash overlap), FIX-2+FIX-4 achieves 80.69% accuracy versus 77.64% (L1), 77.08% (S1), and 75.14% (TALE). Source-stratified 95% bootstrap confidence intervals are strictly positive for all four pairwise comparisons, including versus the source-local best external (CI lower bound +1.11 pp). We also report a stronger pooled-answer ensemble baseline over frontier, L1, S1, and TALE; FIX-2+FIX-4 remains ahead by point estimate on Final-300 and Aggregate-720, but that comparison is not statistically separated. All claims are explicitly scoped to the tested provider and benchmark; we make no universal generalization claims.

We also document five non-promoted prototypes (FIX-5 through FIX-9) evaluated against the same criteria, following a protocol that separates exploratory development from promoted evidence. These are reported as negative or insufficient results in Section 5.1 and Table 10.

The remainder of the paper is organized as follows. Section 2 reviews related work on test-time compute scaling, verifier-guided reranking, and cost-aware routing. Section 3.3 specifies FIX-2+FIX-4 precisely, including exact trigger conditions and pseudocode. Sections 4 and 5 describe the evaluation protocol and report results with full confidence intervals. Section 5.1 summarizes ablation variants and non-promoted outcomes. Sections 6–7 discuss implications, limitations, and directions for future work.

2 Related Work

Chain-of-thought and self-consistency.

Chain-of-thought (CoT) prompting [5] and its zero-shot variant [6] demonstrated that step-by-step reasoning elicits substantially stronger performance from large language models on mathematical and logical tasks. Building on CoT, Wang et al. [3] introduced self-consistency, in which multiple reasoning paths are sampled and the most frequent final answer is returned. Self-consistency is a direct ancestor of the exploration strategy in this work: the frontier method accumulates a pool of candidate answers via diverse sampling before applying a selection policy. The key difference from self-consistency is that our selection policy is conditioned on logged execution metadata—specifically failure-trace signals from the frontier run—rather than raw majority vote across all samples.

Test-time compute scaling.

Recent work has shown that scaling inference-time computation can yield accuracy gains comparable to scaling model parameters [2]. Budget-forcing [7] operationalises this by extending or truncating the reasoning trace to a target token count, encouraging the model to use its full compute allotment. Wu et al. [8] and related work study cost–performance trade-offs across test-time strategies (for example, sampling, voting, and search variants) under fixed compute budgets. The present work occupies a complementary niche: given candidate answers generated under a common per-method budget cap (budget = 6 in our setting), we ask how to select the best answer from the available pool without additional generation.

Tree-structured and search-based reasoning.

Tree of Thoughts [9] and similar frameworks [10] extend CoT to multi-branch deliberate search, where the model evaluates partial solutions at each step. These approaches improve coverage of the solution space at the cost of increased inference computation. The frontier method used as our baseline can be viewed as a lightweight variant: it generates candidates via program-aided execution [4], accumulates a diverse pool under a budget constraint, and selects a final answer via a fixed rule. The failure modes we identify—structurally weak single-branch explorations and frontier-external disagreements—arise specifically because not all budget allocations produce equally reliable candidates, motivating the gating rules in FIX-2 and FIX-4.

Verifier-guided reranking and outcome reward models.

A complementary line of work trains separate verifier or reward models to rerank candidate solutions. Cobbe et al. [1] showed that a trained outcome verifier substantially improves answer selection on the GSM8K benchmark. Process reward models [11] extend this to step-level verification, providing finer-grained feedback. In our setting, a SetFit verifier [12] was trained during the RelationReady phase and showed positive within-method reranking signal; however, cross-method routing was method-entangled and not promoted. The gating rules studied in this paper avoid learned verifier components entirely: they rely only on runtime metadata that is logged as a side-effect of the frontier run.

Cost-aware routing and cascades.

FrugalGPT [13] and related cascade approaches [14] route queries to cheaper models or early-exit points when a confidence threshold is met. These methods reduce overall inference cost by skipping expensive models on "easy" examples. Our setting differs in that each candidate-producing method is evaluated under the same budget contract: no compared method receives a larger per-example call cap, and the selection policy itself issues no calls. The allocation question is therefore not which model to call, but which answer to surface from an already-generated candidate pool, given only gold-free execution signals.

Prompt-budget strategies.

Several approaches directly manipulate the prompt or generation parameters to control inference cost. TALE-style prompt budgeting, used as an external baseline in this paper, allocates a target token count in the prompt and constrains generation length. Such approaches are orthogonal to our method: they control the generation phase, while FIX-2+FIX-4 acts post-generation on already-produced candidates.

Adaptive computation.

Adaptive computation in neural networks—from early exits [15] to mixture-of-experts gating—dynamically allocates compute based on input difficulty. Our failure-trace gating policy can be seen as an adaptive selection layer applied at the output of the frontier generator: it redirects the final answer for a predicted subset of "weak" outputs without requiring additional forward passes.

Positioning of this work.

The present paper contributes a simple, deterministic, gold-free gating policy evaluated under strict reproducibility constraints: exact rule definitions, matched budgets, disjoint evaluation splits, and explicit confidence-interval reporting. Unlike verifier-based approaches, the method requires no additional model training. Unlike search-based generation methods, it adds no inference calls after candidate answers and frontier metadata have been produced. The principal contribution is an empirical demonstration that two lightweight failure-trace signals—the override-reason metadata field and external-consensus agreement—can recover a meaningful fraction of cases where the frontier method's output is unreliable, yielding statistically supported accuracy gains on the tested Cohere setting.

3 Problem Setup and Method

3.1 Problem Setup

We consider method selection under a fixed per-method budget parameter (budget = 6) on Cohere command-r-plus-08-2024 runs over `openai/gsm8k` examples. GSM8K, a grade-school math benchmark, was introduced by Cobbe et al. [1] and provides 1,319 test problems with numerical final answers that are amenable to exact-match evaluation. For each example, candidate-producing methods produce final-answer candidates under the same provider, model, and budget cap. The objective is to maximize exact-match accuracy without using gold labels at runtime.

The frontier method is a direct-reserve semantic frontier generator. At a high level, it allocates the fixed budget across a small set of direct-reserve attempts and frontier exploration branches, canonicalizes their final answers into answer groups, and selects a default frontier answer from the resulting candidate pool. During this execution it records runtime trace metadata: answer-group counts, support for the selected frontier group, whether the direct-reserve answer agrees with the frontier answer, and an `override_reason` field summarizing why the frontier selector committed to its final answer. The allocator proposed in this paper treats those metadata as diagnostics, not as correctness labels.

Two `override_reason` values are central. The value `single_weak_frontier_branch` is emitted when the selected frontier answer is supported by a narrow or weak frontier trace, which suggests limited exploration of alternative answer groups in this evaluated setting. The value `direct_frontier_agree` is emitted when the direct-reserve and frontier answers agree internally. Both fields are produced before any comparison to the gold answer; they are therefore runtime-visible and gold-free.

Evaluation is reported on:

- final300 (seed 71),
- aggregate720 disjoint primary set (seeds 41, 61, 71),
- optional sensitivity set (+100; labeled sensitivity-only).

3.2 Terminology

The following terms are used consistently throughout the paper. Where helpful for readability, we use short paper-facing names and keep the corresponding implementation identifiers in Table 16 for reproducibility.

Candidate pool

The set of final answers produced by the available methods (frontier and external baselines) for a given example. Candidate-pool construction is separate from the downstream selection policy.

Frontier method
: The primary, computationally intensive answer selected by the direct-reserve semantic frontier policy (`direct_reserve_semantic_frontier_v2`), which serves as the default output before any policy gate fires.

External baseline
: Any of the three independently-sourced single-model methods used as reference signals: L1, S1, or TALE (defined below). External baselines provide gold-free consensus signals without access to gold labels.

L1 An external method that selects the answer with the highest support count across sampled model outputs.

S1 An external method that applies sequential budget-forcing to elicit a final answer within the allocated budget.

TALE An external method that uses prompt-level budget cues to guide answer selection.

Logical API call
: A single provider inference request counted against a candidate-producing method's budget.

Budget The integer upper bound $B = 6$ on logical API calls per example for each candidate-producing method, applied identically to all compared methods (matched-budget condition).

Matched budget
: All compared candidate-producing methods operate under the same per-method budget cap B, ensuring that accuracy differences are not confounded by unequal per-method computational limits. If a deployment generates frontier, L1, S1, and TALE outputs from scratch, its total wall-clock and API cost depends on which producers are executed.

Postrun replay
: Offline re-evaluation of a previously generated run artifact using a policy rule without issuing new provider API calls. Postrun replay is used to compute policy-specific accuracy from existing logs.

Gold-free runtime feature
: Any input to a policy decision rule that is derived solely from model outputs, run metadata, or structural properties of the candidate pool—never from gold labels, exact-match signals, or correctness indicators.

Promotion-grade evidence
: Evidence collected on held-out evaluation sets that were not used during rule development, meeting the disjointness, seed-independence, and aggregate scale criteria documented in Section 4.

Provider-specific evidence
: Results obtained from a particular provider–model pairing (here, Cohere command-r-plus-08-2024). Claims are explicitly scoped to this setting and do not generalize beyond it without additional replication.

3.3 Method

The promoted method, which we term the **failure-trace-guided allocator** (FIX-2+FIX-4), is an answer-selection layer applied *after* candidate generation (Figure 1). It does not generate new candidates and does not issue additional inference calls: it selects a final answer from already-produced frontier and external outputs generated under the same per-method budget cap. Its two gates—the *weak-frontier trigger* and the *direct-agreement trigger*—fire in strict priority order; at most one gate acts per example. All runtime decisions are gold-free: no `gold_answer`, `exact_match`, or correctness label is accessed.

Throughout the paper we use short paper-facing names for readability. The combined FIX-2+FIX-4 policy is the *failure-trace-guided allocator*. Its two internal conditions are the *weak-frontier trigger* and the *direct-agreement trigger*. The three external baselines are abbreviated *L1*, *S1*, and *TALE*; the best-performing external on a given evaluation split is the *best external*. Implementation-name mapping is retained in Table 16 for reproducibility.

At each example, the allocator consumes three classes of runtime-legal inputs: the frontier answer candidate, the three evaluated external-baseline answers, and failure-trace metadata emitted during normal frontier execution. The allocator output is a single final answer plus an explicit policy outcome (`fix2`, `fix4`, or frontier retained), so every decision can be audited after the run using only logged runtime fields. This interface is intentionally narrow: the promoted policy does not depend on learned scores or post-hoc labels.

Conceptually, the allocator uses three abstract predicates. $\textsc{WeakFrontier}(x)$ is true when the frontier trace indicates low support or a single-branch selection. $\textsc{ExternalMajority}(x)$ returns an external answer when at least two of L1, S1, and TALE agree, with a deterministic fallback only when all three differ. $\textsc{ExternalConsensusDisagree}(x)$ is true when L1, S1, and TALE unanimously agree with one another and disagree with the frontier answer. The raw log strings in Algorithm 1 and Table 6 are implementation realizations of these predicates, not scientific assumptions tied to a private script.

The weak-frontier trigger is motivated by a structural reliability pattern. When the frontier trace indicates a single weak branch (for example, low-depth termination in the logged metadata), the resulting candidate is treated as evidence of under-exploration in this evaluated setting. In those cases, routing to an external majority answer provides an independent cross-policy signal without spending extra compute.

The direct-agreement trigger addresses a complementary pattern. If the frontier and its reserve agree internally, but all three external methods unanimously select a different answer, this unanimous disagreement is treated as high-confidence evidence that the frontier selection may be wrong under the matched-budget protocol. Partial external agreement (for example, only one or two externals disagreeing) is intentionally not used in the promoted policy, because it is more ambiguous in this setting and did not provide promotion-grade evidence.

The control flow is a strict precedence chain (Figure 1, Table 6). The weak-frontier trigger is checked first; when it fires, the policy routes to an external majority answer and stops. This priority reflects the interpretation of the weak-frontier trace as a generation-quality warning: when exploration itself appears narrow, the policy does not also ask whether the frontier was internally self-consistent. If it does not fire, the policy checks the direct-agreement trigger; when that condition fires, the policy routes to the external unanimity answer. If neither trigger fires, the frontier answer is retained. The default therefore preserves the original frontier method on all examples where neither high-confidence failure trace is present. Conflict handling is deterministic: at most one gate can act, and external majority ties are resolved by the fixed priority order documented in Table 6. That fallback is retained for reproducibility rather than claimed as optimal; the sensitivity noted in Table 14 shows that alternative tie orders change Aggregate-720 replay by less than one percentage point.

All trigger features are runtime-legal and gold-free by construction. They are computed only from model outputs and frontier trace metadata that are already visible at inference time, never from `gold_answer`, `exact_match`, example identifiers, or artifact paths. Table 15 gives the complete legality inventory, including allowed inputs and explicitly excluded fields. This legality boundary is central to the method definition, not only a reporting constraint.

Relative to L1, S1, TALE, and the frontier baseline, FIX-2+FIX-4 plays a different role in the stack. L1, S1, TALE, and frontier each provide candidate answers under the same per-method matched budget; the failure-trace-guided allocator then decides whether to keep the frontier answer or override it using external agreement signals. This separation between candidate generation and final allocation is what allows the policy to improve selection quality without additional model calls. Observed gains come from switching the final answer on a small trace-identified subset of examples rather than from increasing inference budget. Table 16 provides the paper-facing to implementation-name mapping.

3.4 Low-Depth Guard (FIX-2)

The Low-Depth Guard detects frontier answers arising from structurally weak search branches and replaces them with an external majority vote. The trigger relies exclusively on fields logged during the frontier run: the override reason string and the candidate-pool group count.

3.5 External-Consensus Fallback (FIX-4)

The External-Consensus Fallback identifies examples where the direct-agreement trigger fires—meaning the frontier agreed with its own internal reserve selection—yet all three external baselines unanimously chose a different answer. When this unanimous disagreement holds, the policy substitutes the external consensus answer.

3.6 Combined Policy Algorithm

Algorithm 1 gives the complete pseudocode using exact runtime field names. Parameter settings and implementation-name mapping are documented in the appendix reproducibility tables.

The exact field-level trigger conditions are also tabulated in Table 6 for reproducibility.

Table 1 Algorithm: Combined Failure-Trace-Guided Allocator (FIX-2+FIX-4). All inputs are gold-free runtime fields. Named constants and name mapping are documented in the appendix reproducibility tables.

Step	Operation
Inputs per example	
	`frontier_answer`, `result_metadata` (from frontier run logs)
	`l1_answer`, `s1_answer`, `tale_answer` (external baseline outputs)
	`override_reason`, `frontier_support` $\in \mathbb{Z}_{\geq 0}$, `candidate_pool_answer_group_count` $\in \mathbb{Z}_{>0}$
Step 1 —Low-Depth Guard (FIX-2)	
1a	**if** `override_reason` contains `"single_weak_frontier_branch"`
	or (`frontier_support` $= 0$ **and** `candidate_pool_answer_group_count` ≥ 2)
	then:
1b	Compute `ext` $\leftarrow$ majority answer among {`l1`, `s1`, `tale`} (at least 2 agree);
	if no majority, use priority order TALE $\succ$ S1 $\succ$ L1
1c	**if** `ext` exists **and** `ext` $\neq$ `frontier_answer`:
	return `ext`, `policy` $\leftarrow$ `fix2` *(guard fires; stop)*
Step 2 —External-Consensus Fallback (FIX-4)	
2a	**if** `override_reason` $=$ `"direct_frontier_agree"`
	and `l1_answer` $=$ `s1_answer` $=$ `tale_answer`
	and `l1_answer` $\neq$ `frontier_answer`
	then:
2b	**return** `l1_answer`, `policy` $\leftarrow$ `fix4` *(consensus fallback fires; stop)*
Step 3 —Default	
3	**return** `frontier_answer`, `policy` $\leftarrow$ `original`
Output	
	Selected answer and `policy` label; no gold labels accessed at any step.

4 Experimental Setup

4.1 Data, model, and protocol

All promoted comparisons are conducted on **Cohere command-r-plus-08-2024** with a fixed per-method budget cap of $B = 6$ logical API calls per example. The dataset is `openai/gsm8k`. Evaluation conditions are matched across all compared candidate-producing methods: every method operates under the same budget cap, provider, and model, so accuracy differences are not confounded by unequal per-method compute.

Matched-budget definition.

A comparison is *matched-budget* if and only if every compared candidate-producing method is subject to the same maximum logical API call count B per example, with the same provider and model. No method in the promoted comparison receives extra calls or a different model. FIX-2+FIX-4 is then applied as a post-generation selection policy over the logged candidate answers and frontier metadata.

4.2 Cost and resource usage

Table 8 reports per-method resource usage measured from logged Aggregate720 runs on Cohere command-r-plus-08-2024. Reported metrics include average logical API calls, average input tokens, average output tokens, and average latency in seconds. Total tokens (input + output) can be derived by summing the respective columns. Budget parameter $B = 6$ is shown for completeness.

FIX-2+FIX-4 adds negligible selection overhead once candidate answers and frontier metadata are available, but the main deployment cost is candidate-pool construction. If a deployment must generate frontier, L1, S1, and TALE outputs from scratch, total wall-clock and API cost depends on which candidate producers are executed and may exceed a single method's $B = 6$ run. In the logged Aggregate720 runs, the frontier method has higher average latency than L1, S1, or TALE. We report cost in logical calls, tokens, and latency; monetary cost depends on provider pricing and is not used as the primary comparison metric. Dollar-normalized cost analysis, latency percentile distributions, and multi-provider token cost benchmarks are identified as future work (Section 6.1).

4.3 Parameter grounding

Table 14 (Appendix) documents the key parameters and design choices used in FIX-2+FIX-4, including their values, roles, and the robustness evidence available. We report parameter grounding and robustness checks rather than a full continuous hyperparameter sweep; a full threshold sweep is left for future work.

4.4 Primary splits and disjointness

The primary aggregate (Aggregate720) is formed from three disjoint sources: seed41 main300, seed61 independent120, seed71 final300. Cross-source overlap checks confirm zero shared example IDs and zero shared question hashes, ensuring that the aggregate evidence is drawn from non-overlapping subpopulations.

4.5 Statistics

We report paired bootstrap confidence intervals [16] ($n = 5{,}000$ resamples) on final300 and source-stratified aggregate confidence intervals on Aggregate720. The bootstrap resample count of 5,000 is standard for stable CI estimation at the sample sizes used. All confidence intervals are two-sided at the 95% level. When shown, the *bootstrap positive-delta fraction* is the fraction of bootstrap resamples in which the FIX-2+FIX-4 delta is positive; it is a descriptive bootstrap summary and should not be interpreted as a classical frequentist p-value.

5 Results and Analysis

Main results are listed in Table 2 and Figure 2. FIX-2+FIX-4 is the promoted policy; all other FIX variants were evaluated and not promoted (see Section 5.1 and Table 10).

Final-300.

On the unbiased 300-example evaluation (seed 71, budget 6), FIX-2+FIX-4 reaches **86.67%** (260/300), versus L1 83.00% (249/300), S1 82.00% (246/300), and TALE 78.33% (235/300). Paired bootstrap 95% confidence intervals (5 000 resamples) for FIX-2+FIX-4 minus each baseline on this split:

- vs. L1: +3.67 pp, 95% CI [+0.33, +7.00], bootstrap positive-delta fraction 0.983
- vs. S1: +4.67 pp, 95% CI [+1.00, +8.33], bootstrap positive-delta fraction 0.993
- vs. TALE: +8.33 pp, 95% CI [+5.00, +12.00], bootstrap positive-delta fraction 1.000

All three CI lower bounds are strictly positive on this split.

Aggregate-720 (primary).

The primary claim rests on the disjoint Aggregate-720 evaluation combining three independent sources (seeds 41, 61, 71; comprising 300, 120, and 300 examples respectively, with no example-id or question-hash overlap). FIX-2+FIX-4 reaches **80.69%** (581/720), versus L1 77.64% (559/720), S1 77.08% (555/720), and TALE 75.14% (541/720). Source-stratified bootstrap 95% confidence intervals (5 000 resamples):

- vs. L1: +3.06 pp, 95% CI [+0.83, +5.42], bootstrap positive-delta fraction 0.995
- vs. S1: +3.61 pp, 95% CI [+1.39, +5.69], bootstrap positive-delta fraction 0.999
- vs. TALE: +5.56 pp, 95% CI [+3.61, +7.50], bootstrap positive-delta fraction 1.000
- vs. source-local best external: +3.06 pp, 95% CI [+1.11, +5.00], bootstrap positive-delta fraction 0.999

All aggregate source-stratified CI lower bounds are strictly above zero. The margin vs. best external exceeds 1 pp even at the lower CI bound.

Pooled ensemble baseline.

We also evaluate a simple pooled answer ensemble over the four already-generated answers: frontier, L1, S1, and TALE (Table 4). The primary pooled baseline uses strict majority vote over the four answers and falls back to the frontier answer for 2–2 ties or all-different cases; this tie-breaker is deterministic and gold-free. On Final-300, this Pooled-4 baseline reaches 84.33% (253/300), while FIX-2+FIX-4 reaches 86.67% (260/300), a +2.33 pp difference with paired bootstrap 95% CI [−0.67, +5.67]. On Aggregate-720, Pooled-4 reaches 79.86% (575/720), while FIX-2+FIX-4 reaches 80.69% (581/720), a +0.83 pp difference with source-stratified 95% CI [−1.11, +2.78]. An external-only L1/S1/TALE majority baseline is even closer: 85.33% on Final-300 and 80.28% on Aggregate-720. Thus, FIX-2+FIX-4 remains ahead by point estimate, but the pooled-ensemble comparison is not statistically separated and is treated as a robustness check rather than a promoted superiority claim. The external fallback order used by FIX-2 is deterministic and fixed for reproducibility; it is not presented as an optimized learned ranker. A replay sensitivity check over alternative external tie orders changes Aggregate-720 FIX-2-only accuracy from 80.00% to 80.42%, with frontier fallback on all-different external ties reaching 80.83%; these variants are reported only as sensitivity because they were not the promoted policy.

Source-by-source breakdown.

Table 3 gives the per-source accuracy for all evaluated methods. Estimates are consistent across sources: FIX-2+FIX-4 leads by point estimate in every individual source split, with seed-41 showing the smallest advantage (83.33% vs. 80.33% for all three baselines) and seed-71 showing the largest (86.67% vs. 83.00% for L1). The seed-61 source is substantially harder in absolute terms: frontier, L1, S1, and TALE all fall to 57.50%, 57.50%, 56.67%, and 54.17%, respectively. The provider/model/budget configuration is unchanged, and the mean question length is only modestly higher than the other sources (48.5 words vs. 46.1 for seed 41 and 44.6 for seed 71). The available evidence therefore points to source-sample difficulty rather than a method-specific run configuration change. FIX-2+FIX-4 still leads the best external on seed 61 by +1.67 pp, and the source-stratified aggregate confidence interval accounts for this source variation.

Win/loss/tie decomposition.

Appendix Table 12 gives the per-example win/loss/tie counts versus each baseline on Final-300. On the Final-300 evaluation, FIX-2+FIX-4 wins on 19 examples vs. L1 (losses: 8, ties: 273), 23 examples vs. S1 (losses: 9, ties: 268), and 27 examples vs. TALE (losses: 2, ties: 271). The extremely low loss counts vs. TALE confirm that the policy's recoveries substantially outweigh its regressions on this evaluation.

Gate-action decomposition.

Table 7 decomposes the promoted policy into gate actions. Across Aggregate-720, FIX-2 fires on 122 examples, FIX-4 fires on 5 examples after FIX-2 does not fire, and no gate fires on 593 examples. The FIX-4 actions are rare but positive in this artifact: before FIX-4, the frontier answer is incorrect on all five; after FIX-4, all five are correct. The same pattern holds on Final-300, where FIX-4 fires on three examples and contributes three wins, zero losses, and zero ties relative to the frontier answer and to FIX-2-only replay.

Sensitivity set (supporting diagnostic only).

Table 3 includes a row for a 100-example set (seed 31, labeled "sensitivity-only") that was used as a development diagnostic during early policy calibration for FIX-1 and FIX-2. Because this set was involved in parameter selection for earlier FIX variants, it is excluded from the primary Aggregate-720 evidence and from the bootstrap confidence intervals reported above. On this set, FIX-2+FIX-4 achieves 82.00%, tied with TALE and ahead of L1 (76.00%) and S1 (77.00%). When the sensitivity set is added to the aggregate ($N = 820$), the positive point-estimate advantage of FIX-2+FIX-4 over all three external baselines is preserved, and the overall picture is unchanged. All promoted accuracy claims in this paper are based on Final-300 and Aggregate-720 only.

Table 2 Main validated results (final300 and aggregate720). Deltas are percentage points vs baselines on aggregate720.

Policy	Final300 Acc.	Final300 c/n	Agg720 Acc.	Agg720 c/n	ΔvsL1	ΔvsS1	ΔvsTALE	Promoted
frontier original	76.67	230/300	75.28	542/720	−2.36	−1.81	+0.14	no
FIX-2	85.67	257/300	80.00	576/720	+2.36	+2.92	+4.86	no
FIX-2+FIX-4	**86.67**	**260/300**	**80.69**	**581/720**	**+3.06**	**+3.61**	**+5.56**	**yes**
FIX-5	78.67	236/300	75.28	542/720	−2.36	−1.81	+0.14	no
Pooled-4 ensemble	84.33	253/300	79.86	575/720	+2.22	+2.78	+4.72	no
External-3 ensemble	85.33	256/300	80.28	578/720	+2.64	+3.19	+5.14	no
L1	83.00	249/300	77.64	559/720	0.00	+0.56	+2.50	no
S1	82.00	246/300	77.08	555/720	−0.56	0.00	+1.94	no
TALE	78.33	235/300	75.14	541/720	−2.50	−1.94	0.00	no
best external	83.00	249/300 (L1)	77.64	559/720 (L1)	0.00	+0.56	+2.50	no

Table 3 Source-by-source accuracy (%) and aggregate context. Sensitivity source (seed 31) is a labeled auxiliary set, not part of promoted Aggregate720 evidence. Bold denotes the best value within each source row (ties allowed).

Source	Seed	N	Frontier	FIX-2	FIX-2+4	L1	S1	TALE	Δ(FIX-2+4 vs best ext.)
main_300_seed41	41	300	81.00	82.67	**83.33**	80.33	80.33	80.33	+3.00
independent_120_seed61	61	120	57.50	**59.17**	**59.17**	57.50	56.67	54.17	+1.67
final_300_seed71	71	300	76.67	85.67	**86.67**	83.00	82.00	78.33	+3.67
sensitivity_100_seed31	31	100	73.00	80.00	**82.00**	76.00	77.00	**82.00**	0.00
Aggregate720 (primary)	41+61+71	720	75.28	80.00	**80.69**	77.64	77.08	75.14	**+3.06**

Table 4 Pooled answer-ensemble baselines computed from the same per-example frontier, L1, S1, and TALE outputs. Pooled-4 uses a strict majority over four answers and falls back to the frontier answer on ties or all-different cases. External-3 uses majority vote over L1/S1/TALE with the fixed, deterministic TALE≻S1≻L1 tie fallback used by the promoted policy; tie sensitivity is summarized in Table 14. Deltas and win/loss/tie counts compare FIX-2+FIX-4 to the pooled baseline.

Split	FIX-2+4	Pooled-4	External-3	Δ vs Pooled-4	W/L/T vs Pooled-4	95% CI
Final-300	260/300 (86.67%)	253/300 (84.33%)	256/300 (85.33%)	+2.33 pp	16/9/275	[−0.67, +5.67]
Aggregate-720	581/720 (80.69%)	575/720 (79.86%)	578/720 (80.28%)	+0.83 pp	28/22/670	[−1.11, +2.78]

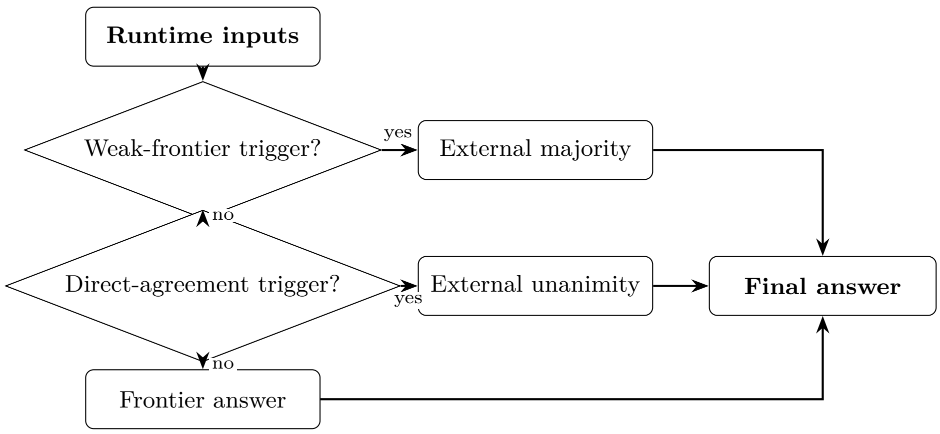
5.1 Ablation Study and Non-Promoted Variants

We evaluate ablation and extension variants to test whether the promoted two-gate allocator can be improved by additional routing, extra-action, clustering, parsing, or pool-miss logic. Table 5 summarizes the evaluated variants from FIX-1 through FIX-9. All variants were evaluated via postrun replay on existing candidate artifacts; no additional inference calls were issued for ablation purposes. None of the variants met promotion criteria beyond FIX-2+FIX-4, and they are reported to delimit the supported claim.

Promoted gate components.

FIX-2 alone achieves 80.00% on Aggregate-720 (+2.36 pp over the frontier baseline), confirming that the Low-Depth Guard carries most of the combined gain. FIX-4 alone achieves 75.00% on final-300—below FIX-2—but contributes an additional +0.69 pp when combined with FIX-2, reflecting its role in recovering cases where FIX-2 does not fire. FIX-3 (calibration on score-absent non-SWFB cases) was evaluated but applied to zero examples in the primary splits, contributing no accuracy change; it is excluded from the combined policy.

Failure-trace-guided allocator



Paper-facing trigger names map to implementation identifiers in Table 16.

Fig. 1 Overview of the failure-trace-guided allocator. The policy first checks the weak-frontier trigger; if it fires, the external majority answer is selected. Otherwise, the policy checks the direct-agreement trigger; if all external baselines unanimously disagree with the frontier answer, the unanimous external answer is selected. If neither trigger fires, the frontier answer is retained. Paper-facing trigger names map to implementation identifiers in Table A6.

Non-promoted variants.

FIX-5 (TALE-default router) produced zero switches on seeds 41 and 61, equalling TALE performance on both splits, and only a marginal net positive switch on seed 71 (+1 example over TALE, final-300 78.67% vs. 78.33%). This outcome is −8.00 pp below FIX-2+FIX-4 on the same split. Note that the Aggregate-720 accuracy for FIX-5 (75.28%) numerically coincides with that of the frontier baseline; this reflects the zero-switch behavior on seeds 41 and 61 (FIX-5 reverts to TALE, and TALE plus the single seed-71 switch happen to sum to the same total as the frontier), not an equivalence by design. The TALE-default design assumed a reliable agreement region that did not materialise consistently across evaluation seeds; see Section 6 for interpretation. FIX-6 (LoVEC extra-action variant) showed a nonnegative pilot signal in the initial 40-case overlapping evaluation, but the independent 80-case Stage-2 relaunch yielded net −1 for extra frontier actions and net −4 for extra TALE retries relative to FIX-2+FIX-4. Early-stage signal that fails to replicate on disjoint data is the standard non-promotion criterion. FIX-7 (cluster-level selector) achieved at most +0.67 pp in the best rule variant on final-300, below the promotion-meaningful threshold. FIX-8 (robust parser/canonicaliser) produced zero accuracy-changing switches on both final-300 and Aggregate-720, indicating that parse-failure residuals are rare in this evaluation setting. FIX-9 (pool-miss detector) found 23 of 24 all-methods-wrong residuals to be generation-limited, with no selector-only remedy available; see the failure analysis in Section 5.2.

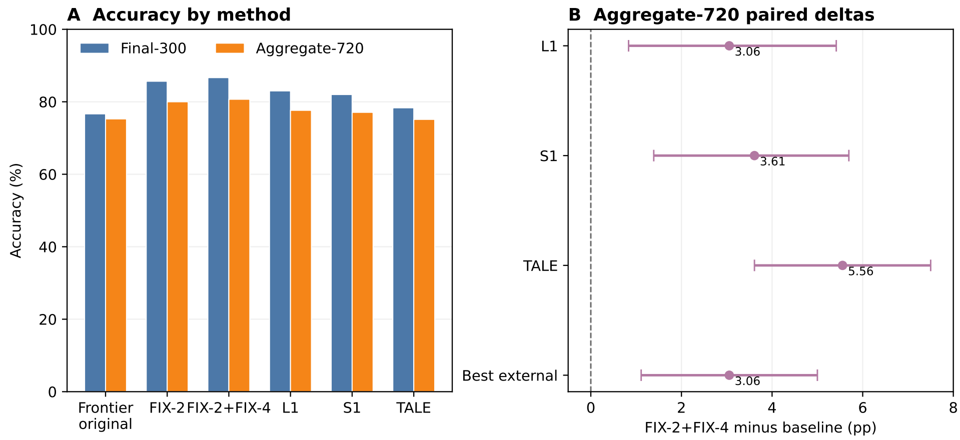


Fig. 2 Main accuracy summary for the promoted policy. Panel A compares FIX-2+FIX-4 with the frontier baseline and external baselines on Final-300 and Aggregate-720. Panel B shows Aggregate-720 paired deltas for FIX-2+FIX-4 versus L1, S1, TALE, and the best external baseline; the zero line marks no improvement. The primary promoted evidence is the disjoint Final-300 and Aggregate-720 evaluation, and the figure is descriptive rather than a new claim.

Table 5 Ablation variants and non-promoted policy outcomes. Bold denotes the strongest promoted-policy result in the main comparison block.

Policy	Target failure mode	Accuracy / delta summary	Promoted
frontier original	baseline reference	final300=76.67%; agg720=75.28%	no
FIX-1 support-aware	tie / present-not-selected	77.00% (delta vs frontier +4.00 pp)	no
FIX-2 low-depth guard	single-weak-frontier-branch / low depth	80.00%	no (superseded by FIX-2+FIX-4)
FIX-3 calibration	score-absent non-SWFB edge cases	73.00% (applied 0 cases)	no
FIX-4 consensus gate	direct-frontier-agree + unanimous external disagreement	75.00% alone; 82.00% with FIX-2	part of FIX-2+FIX-4
FIX-2+FIX-4	low-depth + external-consensus	final300=86.67%; agg720=80.69%	**yes**
FIX-1+2+3+4	combined heuristic stack	81.00%	no
FIX-5 TALE-default router	agreement-region switching	overnight300=80.33%; final300=78.67%	no
FIX-6 LoVEC extra-action	residual errors needing extra action	net vs FIX-2+FIX-4: frontier −1, TALE −4	no
FIX-7 cluster selector	cluster-level reranking / parser guards	best-rule local delta vs FIX-2+FIX-4: +0.67 pp	no
FIX-8 robust parser	parser/canonicalization errors	final300 delta +0.00 pp; agg720 +0.00 pp	no
FIX-9 pool-miss detector	generation-limited residuals	23/24 all-methods wrong generation-likely; 0 rule opportunities	no

5.2 Failure Analysis

Table 9 and Figure 3 classify the 40 residual errors in the Final-300 evaluation by root cause. Three categories emerge. *All-methods-wrong* cases (24/40, 60.0%) are examples where every evaluated method—the frontier generator and all three external baselines—produced an incorrect answer. Because no correct answer appears in the

Table 6 Exact policy rule definitions for reproducibility. Abstract predicates are implemented by gold-free runtime fields only.

Component	Predicate / exact condition	Action	Gold-free rationale
Low-Depth Guard (FIX-2): WEAKFRONTIER	`override_reason` contains `single_weak_frontier_branch` **or** (`frontier_support`= 0 **and** `pool_group_count`≥ 2)	Fallback to external majority vote; if no majority use priority TALE≻S1≻L1; else keep frontier	Uses only runtime metadata and baseline answers; no gold/exact labels
External-Consensus Fallback (FIX-4): EXTERNALCONSENSUSDISAGREE	`override_reason=direct_frontier_agree` **and** `l1=s1=tale` **and** unanimous external ≠ frontier	Replace frontier with the unanimous external answer	Uses only runtime override reason and method answers
Combined FIX-2+FIX-4	If FIX-2 fires: apply FIX-2 and stop; else if FIX-4 fires: apply FIX-4; else keep frontier	Single-fire precedence chain	No correctness labels in trigger chain

Table 7 Gate-action decomposition for the promoted FIX-2+FIX-4 policy. FIX-4 fires only after FIX-2 does not fire. The final columns report correctness before and after FIX-4 on the examples where FIX-4 fires, relative to the original frontier answer.

Split	N	FIX-2 fires	FIX-4 fires	No gate	FIX-4 before	FIX-4 after	FIX-4 W/L/T
seed41	300	38	2	260	0/2	2/2	2/0/0
seed61	120	21	0	99	0/0	0/0	0/0/0
seed71 (Final-300)	300	63	3	234	0/3	3/3	3/0/0
Aggregate-720	720	122	5	593	0/5	5/5	5/0/0

Table 8 Aggregate720 per-method cost and resource summaries from logged runs. Provider: Cohere command-r-plus-08-2024; each candidate-producing method is evaluated with budget cap $B = 6$ (matched-budget condition). Avg total tokens = avg input + avg output. Monetary cost is not reported; it depends on provider pricing (see Section 4.2).

Method	Avg logical calls	Avg input tok.	Avg output tok.	Avg total tok.	Avg latency (s)	Budget
Frontier	2.60	1153.97	229.94	1383.91	13.30	6
L1	1.21	515.87	114.14	630.01	4.11	6
S1	2.62	1110.20	160.50	1270.70	5.15	6
TALE	1.44	612.38	117.90	730.28	4.63	6

candidate pool, no selection policy can recover these cases; they represent a hard ceiling for the current approach. *Parser/canonicalization failures* (12/40, 30.0%) are cases where the correct numerical answer was present in the pool but not extracted or matched, due to format mismatches between the model's output format and the evaluation matcher. *Present-not-selected* cases (4/40, 10.0%) are cases where the correct answer was available in the pool but the FIX-2+FIX-4 rule chain did not surface it.

Within the all-methods-wrong category, the dominant subtype is *multi-step composition traps* (10/24), problems where the error is embedded in intermediate reasoning rather than in the final answer selection step. Near-miss numeric errors (5/24), different-wrong-answer divergence with no correct cluster (4/24), and false consensus (2/24)—where all methods agree on the same incorrect answer—account for most

Table 9 Failure taxonomy for final300 residual errors.

Failure group	Count	Share	Likely fix class
final300 total failures	40	1.000	mixed
all methods wrong	24	0.600	new candidate generation likely
parser / canonicalization issue	12	0.300	parser / canonicalization
present not selected	4	0.100	selector
Subtypes of all-methods-wrong			
multi-step composition trap	10	0.250	selector-only unlikely
near-miss numeric error	5	0.125	selector-only unlikely
different wrong answers, no correct cluster	4	0.100	mixed (rare selector possible)
same wrong answer, false consensus	2	0.050	selector-only unlikely
shared unit/rate trap	2	0.050	selector-only unlikely
quantity selection trap	1	0.025	selector-only unlikely

of the remainder. False-consensus cases are the most resistant to any selection-based intervention: unanimous agreement provides no gating signal when the consensus itself is wrong.

These findings directly inform the scope of FIX-2+FIX-4 and the directions for future work. The all-methods-wrong class (60% of failures) is generation-limited: addressing it requires either stronger base generators or training-time interventions that improve coverage of hard compositional patterns, not better post-hoc selection policies. The parser/canonicalization class (30% of failures) is potentially tractable with improved answer normalization; however, the FIX-8 prototype produced zero accuracy-changing switches in this setting, suggesting the problem is subtler than simple string-matching improvements. The present-not-selected class (10% of failures) represents the only residual opportunity that a selection policy could in principle address without improved candidate generation.

6 Discussion and Limitations

Why failure-trace-guided gating works.

The two gates in FIX-2+FIX-4 target failure modes that are identifiable at runtime without any access to gold labels. The Low-Depth Guard (FIX-2) addresses a structural weakness: when the frontier search terminates with a *single weak branch*—signalled by the `override_reason` field—the resulting answer is drawn from a narrow slice of the candidate space. In such cases, the external majority vote is empirically more reliable. The External-Consensus Fallback (FIX-4) targets a different pattern: when the direct-agreement trigger fires—signalling that the frontier selected its own answer confidently—yet all three independent external baselines unanimously prefer a different answer, that unanimous disagreement constitutes strong evidence that the frontier is wrong. Both signals are available as logged metadata from normal frontier execution; no additional inference or labelling is required.

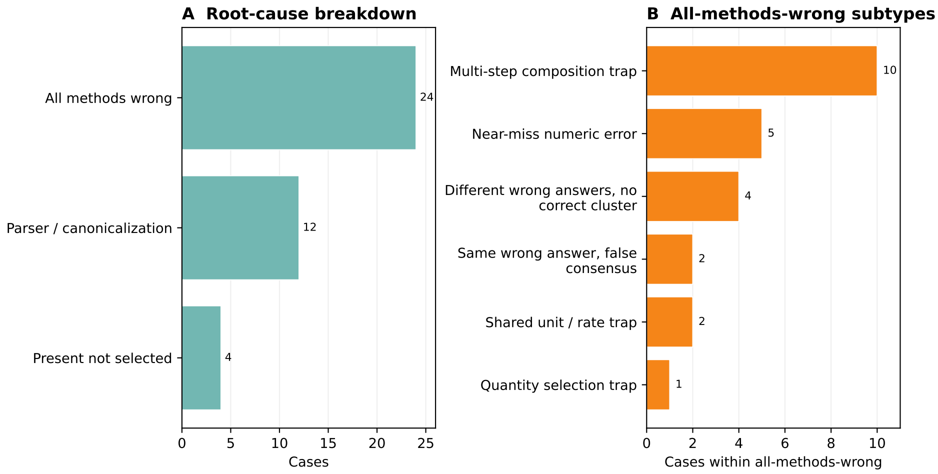


Fig. 3 Descriptive failure analysis for the 40 residual Final-300 errors. Panel A shows the top-level root-cause split; Panel B expands the all-methods-wrong cases into supported subtypes. This is a failure taxonomy, not a new benchmark result, and it is used to frame the residual headroom discussed in the text.

Why the non-promoted variants were abandoned.

FIX-5 (TALE-default router) produced zero switches on seeds 41 and 61, equalling TALE performance exactly on both splits (80.33% and 54.17%, respectively). On seed 71 (Final-300), FIX-5 made one net positive switch over TALE (78.67% vs. 78.33%), a difference of a single example that is negligible compared to the +8.33 pp advantage of FIX-2+FIX-4 over TALE on the same split. The TALE-default design assumed a reliable agreement region that did not materialise consistently across evaluation seeds, making the router effectively inactive in most conditions.

FIX-6/LoVEC proposed collecting extra frontier or TALE-retry actions for predicted failure cases. The first 40-case overlapping pilot showed nonnegative action-value signal, but the independent 80-case Stage-2 relaunch (disjoint from the pilot) yielded net −1 for extra frontier actions and net −4 for extra TALE retries relative to FIX-2+FIX-4, with non-trivial control-slice regressions (8.9% for frontier, 11.1% for TALE retry). The failure to replicate on disjoint data is the key indicator: the pilot signal appears to have been artefact-specific rather than generalisable.

FIX-7 (cluster-level selector) showed only limited positive movement in offline replay, with a best-rule local gain of +0.67 pp on final-300, but it did not provide promotion-grade aggregate evidence or a robust policy improvement and was therefore not promoted. FIX-8 (robust parser/canonicaliser) produced zero accuracy-changing switches on final-300 and aggregate-720. Taken together, these outcomes indicate that the targeted residual patterns are either rare or too weakly actionable in the current evaluation setting.

How aggregate-720 changes the evidence posture.

The most substantive evidence shift in this work is the transition from a single-run 300-example evaluation (seed 41, where the CI lower bound vs. L1 still crossed zero at $[-0.48, +5.71]$) to the disjoint Aggregate-720 evidence (seeds 41, 61, 71). By combining three independently sampled unbiased evaluation splits, the stratified bootstrap CI narrows sufficiently that the lower bound vs. every external baseline is strictly positive. This is not simply a power argument: the source-stratified design controls for seed-specific variance, and the zero-overlap check on example-id and question-hash confirms independence. The result is that the point-estimate advantage of FIX-2+FIX-4 is consistent across all three seeds, which is the primary reason the aggregate CIs no longer include zero.

Evidence scope and generalization.

The promoted findings are scoped to the tested evaluation setting. What this evidence does *not* establish is generalization beyond that scope: whether the same failure-trace signals are informative on other providers, benchmarks, or budget levels is an open empirical question. Existing repository artifacts include earlier MATH-500, AIME, OpenAI/Cohere, and budget-sweep outputs, but these do not contain the same four-method per-example records and frontier trace metadata needed to replay FIX-2+FIX-4 under the current protocol. They are therefore unsuitable as a clean second benchmark/provider/budget result for this manuscript without new matched-budget generation. The most efficient next replication would be a second GSM8K provider run or a same-provider second benchmark with the identical frontier, L1, S1, and TALE logging contract. Section 6.1 enumerates the scope boundaries explicitly.

6.1 Limitations

We identify the following scope boundaries and residual risks explicitly.

Provider-specific scope.

All promoted evidence is obtained on Cohere command-r-plus-08-2024. The method is not claimed to generalize to other providers, model families, or API versions without independent replication. Provider-specific evidence is the appropriate evidential category for this submission.

Benchmark scope.

Results are scoped to GSM8K (`openai/gsm8k`) under the matched-budget protocol described in Section 4. We make no universal accuracy claim beyond this dataset and task family. Extension to other reasoning benchmarks or open-ended tasks is left for future work.

Moderate margin.

The primary gain of FIX-2+FIX-4 over the best external baseline on Aggregate720 is +3.06 percentage points (95% CI $[+1.11, +5.00]$). While the confidence interval lower bound is positive, the margin is moderate in absolute terms. We do not claim large or transformative improvements.

Pooled-ensemble comparison.

Against a pooled four-answer ensemble over frontier, L1, S1, and TALE, the FIX-2+FIX-4 advantage is smaller: +0.83 pp on Aggregate720 with a confidence interval that includes zero. The pooled comparison is therefore reported as a robustness baseline, not as statistically separated evidence of superiority.

Cost metric scope.

We report cost in logical API calls, tokens, and latency; monetary cost depends on provider pricing and is not used as the primary comparison metric. The matched-budget claim is a per-method evaluation contract, not a claim that generating the full frontier/L1/S1/TALE candidate pool costs the same as one method run. Dollar-normalized cost analysis and latency percentile distributions across all run batches are incomplete and identified as future work.

Tie-breaker sensitivity.

The TALE$\succ$S1$\succ$L1 fallback used when all three external answers differ is deterministic and gold-free, but it is not claimed to be optimal. Alternative tie orders move Aggregate720 replay by less than one percentage point in the available artifacts. A fully optimized tie policy would require separate validation and is outside the promoted claim.

Residual pool-miss failures.

A residual failure class exists in which all methods—frontier and external baselines—produce incorrect answers for the same example. The proposed policy cannot recover from this class, as no correct answer is present in the candidate pool. The fraction of such cases is documented in Table 9 but no remedy has been developed.

Non-promoted prototypes.

Follow-up prototypes (FIX-7, FIX-8, FIX-9) were explored during development but did not meet promotion-grade evidence criteria. They are reported as negative results in Table 10 and must not be interpreted as deployment-ready improvements. Only FIX-2+FIX-4 is promoted.

7 Conclusion

We presented FIX-2+FIX-4, a deterministic gold-free policy gating method for matched-budget LLM reasoning. On a disjoint 720-example aggregate evaluation (three independent seeds, zero overlap), FIX-2+FIX-4 achieves 80.69% (581/720) versus L1 77.64%, S1 77.08%, and TALE 75.14%. Source-stratified 95% bootstrap confidence intervals are strictly positive for all four baseline comparisons, with the lower bound vs. best external at +1.11 pp. On the Final-300 evaluation (seed 71, budget

Table 10 Negative results and non-promotion outcomes.

Prototype	Result	Implication
FIX-5 TALE-default router	Failed as preferred policy on larger unbiased validation; no reliable transfer to promoted setting.	Keep FIX-2+FIX-4 as promoted policy; do not route by TALE-default in deployment claim.
FIX-6 LoVEC extra-action	Independent stage2 relaunch negative net vs FIX-2+FIX-4 (extra_frontier −1, extra_tale −4).	No accuracy-changing LoVEC policy promotion.
FIX-7 cluster selector	Weak final300 gain (+0.67 pp) with non-promotion decision.	Prototype only; insufficient for promotion.
FIX-8 robust parser/canonicalizer	0 recoveries and 0 aggregate gain (final300 and aggregate720 deltas both 0).	No parser-driven policy change justified.
FIX-9 pool-miss detector audit	23/24 cases generation-likely; selector-only opportunity limited.	Residual error frontier is mostly candidate-generation limited under current logs.

6), FIX-2+FIX-4 reaches 86.67% (260/300), compared to 83.00% for L1, 82.00% for S1, and 78.33% for TALE.

The policy consists of two rule-based gates applied in strict priority order. The Low-Depth Guard (FIX-2) detects structurally weak frontier search branches using the logged `override_reason` field and substitutes the external majority vote. The External-Consensus Fallback (FIX-4) identifies cases where all three external baselines unanimously disagree with the frontier's selection, overriding to the consensus answer. Both gates are gold-free: no correctness labels, example identifiers, or artifact paths are accessed at runtime. The rules, trigger conditions, and exact field names are fully specified in Table 6 and Algorithm 1, with reproducibility details documented in the appendix and supplementary artifact.

Further accuracy gains in this setting will most likely require advances in candidate generation rather than selection policy. Failure analysis of the 40 residual errors on Final-300 shows that 24 (60%) are *all-methods-wrong* cases in which no evaluated method produced the correct answer; the dominant subtype is multi-step composition traps (10/24) where the error is embedded in intermediate reasoning rather than in the final answer selection step. Addressing this residual category requires either stronger base generators or training-time interventions that improve coverage of hard compositional patterns.

Acknowledgements

The author thanks Professor Ioannis Koutis for his support, guidance, and helpful discussions during this work. The author is also deeply grateful to his mother for her continuous support and encouragement. The author acknowledges Cohere for providing research API credits that supported the model-inference experiments in this study.

References

[1] Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., Hesse, C., Schulman, J.: Training verifiers to solve math word problems. arXiv preprint arXiv:2110.14168 (2021)

[2] Snell, C., Lee, J., Xu, K., Kumar, A.: Scaling LLM test-time compute optimally can be more effective than scaling parameters for reasoning. In: The Thirteenth International Conference on Learning Representations (2025). Oral. https://openreview.net/forum?id=4FWAwZtd2n

[3] Wang, X., Wei, J., Schuurmans, D., Le, Q.V., Chi, E.H., Narang, S., Chowdhery, A., Zhou, D.: Self-consistency improves chain of thought reasoning in language models. In: International Conference on Learning Representations (2023)

[4] Gao, L., Madaan, A., Zhou, S., Alon, U., Liu, P., Yang, Y., Carbonell, J., Neubig, G.: PAL: Program-aided language models. In: International Conference on Machine Learning, pp. 10764–10799 (2023). PMLR

[5] Wei, J., Wang, X., Schuurmans, D., Bosma, M., Ichter, B., Xia, F., Chi, E.H., Le, Q.V., Zhou, D.: Chain-of-thought prompting elicits reasoning in large language models. In: Advances in Neural Information Processing Systems, vol. 35, pp. 24824–24837 (2022)

[6] Kojima, T., Gu, S.S., Reid, M., Matsuo, Y., Iwasawa, Y.: Large language models are zero-shot reasoners. In: Advances in Neural Information Processing Systems, vol. 35, pp. 22199–22213 (2022)

[7] Muennighoff, N., Yang, Z., Shi, W., Li, X.L., Fei-Fei, L., Hajishirzi, H., Zettlemoyer, L., Liang, P., Candès, E., Hashimoto, T.: s1: Simple test-time scaling. In: Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing, pp. 20275–20321. Association for Computational Linguistics, Suzhou, China (2025). https://doi.org/10.18653/v1/2025.emnlp-main.1025 . https://aclanthology.org/2025.emnlp-main.1025/

[8] Wu, Y., Sun, Z., Li, S., Welleck, S., Yang, Y.: Inference scaling laws: An empirical analysis of compute-optimal inference for problem-solving with language models. arXiv preprint arXiv:2408.00724 (2024)

[9] Yao, S., Yu, D., Zhao, J., Shafran, I., Griffiths, T.L., Cao, Y., Narasimhan, K.: Tree of thoughts: Deliberate problem solving with large language models. In: Advances in Neural Information Processing Systems, vol. 36 (2023)

[10] Besta, M., Blach, N., Kubicek, A., Gerstenberger, R., Podstawski, M., Gianinazzi, L., Gajda, J., Lehmann, T., Niewiadomski, H., Nyczyk, P., Hoefler, T.: Graph of thoughts: Solving elaborate problems with large language models. In: Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence, vol. 38, pp.

17682–17690 (2024). https://doi.org/10.1609/aaai.v38i16.29720

[11] Lightman, H., Kosaraju, V., Burda, Y., Edwards, H., Baker, B., Lee, T., Leike, J., Schulman, J., Sutskever, I., Cobbe, K.: Let's verify step by step. In: International Conference on Learning Representations (2024)

[12] Tunstall, L., Reimers, N., Jo, U.E.S., Bester, L., Eisenschlos, J., Marcelino, V., Wolf, T.: Efficient few-shot learning without prompts. In: Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing: Industry Track, pp. 1–7 (2022)

[13] Chen, L., Zaharia, M., Zou, J.: FrugalGPT: How to use large language models while reducing cost and improving performance. Transactions on Machine Learning Research (2024). ISSN 2835-8856

[14] Dohan, D., Xu, W., Lewkowycz, A., Austin, J., Bieber, D., Gontijo Lopes, R., Wu, Y., Michalewski, H., Saurous, R.A., Sohl-dickstein, J., Murphy, K., Sutton, C.: Language model cascades. arXiv preprint arXiv:2207.10342 (2022)

[15] Graves, A.: Adaptive computation time for recurrent neural networks. arXiv preprint arXiv:1603.08983 (2016)

[16] Efron, B., Tibshirani, R.J.: An Introduction to the Bootstrap. Chapman & Hall/CRC, New York (1993)

A Reproducibility and Supplementary Material

B Parameter Grounding and Sensitivity

C Runtime Feature Legality

D Paper-Facing Name Mapping

E Algorithm Listing

The promoted policy is presented inline as Algorithm 1 in Section 3.6. For the full executable implementation, see experiments/support_aware_selector.py and the supplementary reproducibility artifact at outputs/machine_learning_journal_result_package_20260520_20260520T035127Z/algorithm_fix24_pseudocode.md.

Data Availability

The code and processed evaluation artifacts are maintained in a private development workspace during manuscript preparation. A cleaned, versioned artifact release will be made public upon acceptance, preferably with a persistent identifier such as a Zenodo

Table 11 Reproducibility checklist.

Item	Status	Value	Artifact/reference
Dataset	complete	`openai/gsm8k` primary subsets (300+120+300)	manifests + `aggregate_with_prior_validation.csv`
Provider / model	complete	Cohere / command-r-plus-08-2024	`validation_manifest.json`
Seeds	complete	41, 61, 71 (primary); 31 sensitivity	`aggregate_with_prior_validation.csv`
Budget	complete	$B = 6$ matched across all methods	manifests + per-example rows
Output roots	complete	major roots enumerated	supplementary reproducibility artifact + `docs/LATEST_RESULTS_AND_CLAIMS.md`
Disjointness proofs	complete	source overlaps = 0	`final_postrun_metrics.json` checks
Dry-run logs	complete	dry-run plans present for all validation runs	`dry_run_call_plan.log`
Postrun outputs	complete	final postrun metrics and CI tables present	`final_fix24_all_external_postrun` root
Bootstrap method	complete	5000 resamples; pooled + source-stratified summaries	`bootstrap_ci_summary.csv`
Policy code files	complete	`support_aware_selector.py`, `cluster_answer_selector.py`, `robust_answer_parser.py`	`experiments/`
Tests	partial	policy/parser tests present; not all scripts formalized as tests	`tests/`
Non-promoted prototypes	complete	FIX-5/6/7/8/9 documented with negative or insufficient evidence	Table 10 + supplementary documentation
Rerun requirements	partial	API credentials and provider availability required for live rerun	validation scripts/manifests

DOI. A clean supplementary reproducibility package is included with the submission, and reviewer-access copies of the processed artifacts can be provided upon request during review. The GSM8K benchmark questions used for evaluation are available from the original source [1]. No proprietary or restricted data were used.

Code Availability

The complete implementation of FIX-2+FIX-4 is provided in experiments/support_aware_selector.py within a private development workspace. A cleaned, versioned public release will be made available upon acceptance, preferably with a persistent identifier such as a Zenodo DOI. Evaluation driver scripts are located under scripts/. Reviewer-access processed artifacts or supplementary files can be provided upon request, but no API keys or other private internal artifacts will be released.

Table 12 Win/loss/tie decomposition for FIX-2+FIX-4 versus each external baseline on Final-300 (seed 71, budget 6, $n = 300$). A win is a case where FIX-2+FIX-4 is correct and the baseline is not; a loss is the reverse; a tie is when both policies agree on correctness. The low loss counts confirm that FIX-2+FIX-4 recoveries substantially outweigh regressions, particularly vs. TALE.

Comparison	Wins	Losses	Ties
FIX-2+FIX-4 vs L1	19	8	273
FIX-2+FIX-4 vs S1	23	9	268
FIX-2+FIX-4 vs TALE	27	2	271

Table 13 Claim language guidance for manuscript writing.

Type	Claim statement
Safe	FIX-2+FIX-4 outperforms L1/S1/TALE and best-external on aggregate720 (matched budget), with positive source-stratified bootstrap CI lower bounds for all baseline comparisons.
Safe	FIX-2+FIX-4 is ahead of pooled-answer ensemble baselines by point estimate on Final-300 and Aggregate-720, but the confidence intervals include zero.
Safe	Failure-trace-guided gold-free rules recover low-depth and external-consensus failure modes.
Safe	FIX-5/6/7/8/9 are not promoted due to insufficient or negative evidence.
Unsafe	Universal superiority over all reasoning methods.
Unsafe	Statistically separated superiority over pooled-answer ensemble baselines.
Unsafe	Universal gains across all math datasets/providers.
Unsafe	Promoted gains from FIX-7/FIX-8/FIX-9.
Unsafe	Dollar/latency savings without direct measurement.

Funding

This work was supported by Cohere research API credits. No other specific grant funding was received from any funding agency in the public, commercial, or not-for-profit sectors.

Competing Interests

The authors declare no competing interests.

Use of AI Tools

The author used ChatGPT, Cursor, Claude, GitHub Copilot, Cohere models, and Perplexity AI during parts of the coding, literature-search, manuscript-drafting, editing, and artifact-preparation workflow. These tools were used for assistance with code navigation, implementation support, language editing, reference discovery, and drafting support. All scientific claims, experimental design choices, code changes, analyses, reported results, citations, and final manuscript text were reviewed and approved by the author, who takes full responsibility for the content of the paper.

Author Contributions

Soroush Vahidi: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review and editing, Visualization.

Table 14 Parameter grounding for FIX-2+FIX-4 and the evaluation protocol. We report parameter grounding and available robustness checks rather than a full continuous hyperparameter sweep; a full threshold sweep is left for future work. All values are fixed and applied identically across all evaluated splits.

Parameter / Design choice *(value used)*	Role, justification	Robustness evidence available	Remaining limitation
Budget ($B = 6$)	Hard cap on logical API calls per example, ensuring matched-budget comparison. Fixed by provider run protocol; all methods use the same B.	Sensitivity source (+100 examples) included as separate labeled set; B held constant.	No multi-B sweep reported.
Low-depth guard: `single_weak_frontier_branch` *(string match; implementation constant)*	Identifies frontier answers from structurally weak search paths. Derived from failure-trace analysis of frontier run logs (`support_aware_selector.py`).	Trigger frequency and recovery/regression counts available in postrun artifacts.	No threshold variation sweep in manuscript.
Low-depth guard: `frontier_support`$=0 \wedge$ `pool_group_count`≥ 2 *(integer thresholds: 0 and 2)*	Detects zero-support frontier outputs when the pool contains ≥ 2 distinct answer groups. Conservative: requires complete absence of support plus pool diversity.	Condition derived from offline evaluation review; trigger counts in supplementary artifacts.	No continuous threshold sweep.
External-consensus condition (FIX-4): `l1`=`s1`=`tale` $\wedge$ all $\neq$ frontier *(unanimity gate)*	Identifies unanimous external agreement diverging from frontier. Unanimity is a strong, conservative signal; partial agreement is not acted on.	Trigger frequency available from postrun replay artifacts.	No partial-consensus variant evaluated.
Bootstrap resamples *($n = 5{,}000$)*	Controls Monte Carlo variance in paired bootstrap CI estimation. Standard practice for stable 95% CIs at $n \leq 720$.	Single-seed results show consistent sign agreement.	No resample-count sensitivity sweep reported.
Evaluation seeds / splits *(seeds 41, 61, 71)*	Define disjoint aggregate and source-stratified analysis. Seeds span independent run batches; disjointness verified by example ID and question hash.	Source-stratified CIs show consistent positive sign for all baselines.	Single task/provider; no cross-dataset seed check.
Disjointness check *(overlap = 0)*	Prevents data leakage between primary sources. Verified programmatically on all three source sets.	Confirmed in postrun evaluation artifacts; overlap count = 0.	Within GSM8K only; no cross-dataset overlap check.
Provider / model *(Cohere command-r-plus-08-2024)*	Inference backend for all runs. Chosen for API availability and run stability during the experiment period.	All compared methods use the same provider/model.	Provider-specific; generalization not established.
External tie fallback *(TALE≻S1≻L1)*	Deterministic gold-free fallback when L1, S1, and TALE all disagree. Retained as the fixed promoted-policy convention rather than optimized on Aggregate720.	Alternative FIX-2-only replays on Aggregate720 give 80.14% for S1-first and 80.42% for L1-first; frontier fallback on all-different external ties gives 80.83%.	Tie order is not separately validated as an optimal policy.

Table 15 Runtime feature legality for FIX-2+FIX-4. All gating inputs are logged as side-effects of normal frontier execution and are available at inference time without any access to gold answers, correctness labels, example IDs, or artifact paths.

Feature	Source	Gold-free?	Why it is allowed
`override_reason`	Frontier run metadata	Yes	Logged search-state signal; e.g., `"single_weak_frontier_branch"` or `"direct_frontier_agree"`; not a correctness label.
`frontier_support`	Frontier run metadata	Yes	Integer support count from the frontier search; derived from pool structure, not correctness.
`candidate_pool_answer_group_count`	Frontier run metadata	Yes	Number of distinct answer groups in the pool; reflects diversity, not correctness.
`frontier_answer`	Frontier generator output	Yes	Frontier candidate selected before gating; proposed output only.
`l1_answer`	External L1 baseline output	Yes	L1 inference output, not the ground-truth label.
`s1_answer`	External S1 baseline output	Yes	S1 budget-forcing output only.
`tale_answer`	External TALE baseline output	Yes	TALE prompt-budgeting output only.
Excluded at runtime	N/A	N/A	`gold_answer`, `exact_match`, `example_id`, and artifact paths are never used in routing.

Table 16 Paper-facing name mapping. Short names used throughout the manuscript and their implementation identifiers, roles, runtime availability, and gold-free status. Raw identifiers are retained in Algorithm 1, Table 6, and Table 15.

Paper-facing name	Implementation identifier	Role	Used at runtime?	Gold-free?
failure-trace-guided allocator	`FIX-2+FIX-4`	Promoted two-gate policy	Yes	Yes
weak-frontier trigger	`single_weak_frontier_branch`	Low-Depth Guard condition; `override_reason` value	Yes	Yes
direct-agreement trigger	`direct_frontier_agree`	External-Consensus Fallback condition; `override_reason` value	Yes	Yes
L1	`external_l1_max`	External baseline; L1-max answer selector	Yes	Yes
S1	`external_s1_budget_forcing`	External baseline; S1 budget-forcing method	Yes	Yes
TALE	`external_tale_prompt_budgeting`	External baseline; TALE prompt-budgeting method	Yes	Yes
best external	best external comparator	Highest-accuracy external baseline on a given split	No	Yes